Week 3:
Overpopulation and *Overconsumption*

WI 1103 Introduction to Principle of Sustainability



Part 3.1:
Planetary Boundary and Environmental Support Capacity

WI 1103 Introduction to Principle of Sustainability

Take home message

"The Future We Want" (Rio+ 20s)

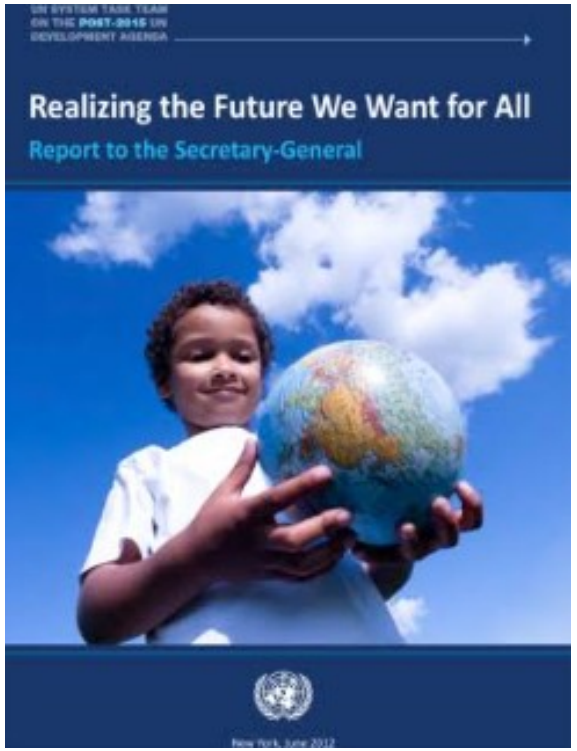
'Realizing The Future We Want for All (2019)

- Since Rio 1992, the world has agreed to navigate development with the compass of sustainability—economic progress must be in harmony with social justice and environmental preservation.

- Human activities must continue to meet the limits of the planet (planetary boundaries / safe operating space).

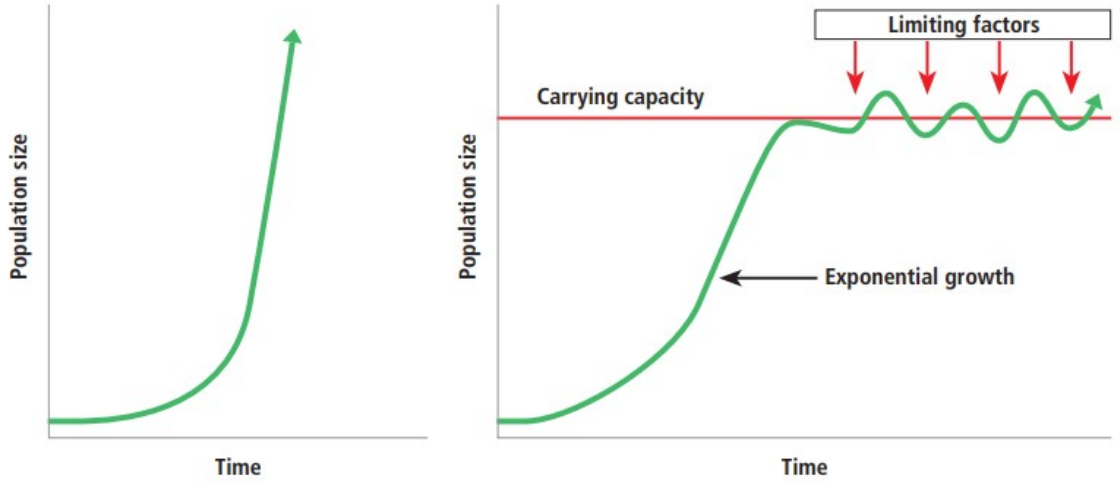
*Based on previous lectures, 7 out of 9 planetary boundaries have been breached.





The concept of environmental carrying capacity

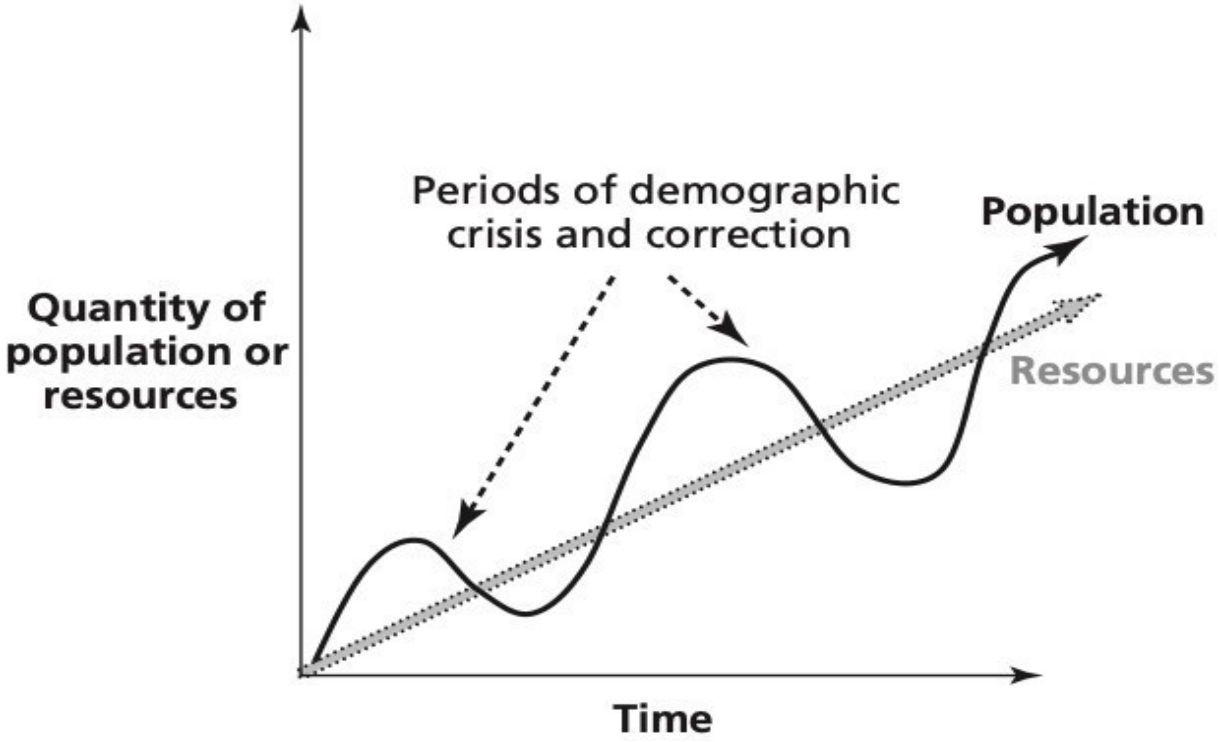
- Naturally, there is always a limit to population growth. Limiting factors include space, abiotic factors, and biotic factors (e.g.: disease).
- Limiting factors determine the carrying capacity of an area, which is the maximum number of individuals of a species that can be sustainably supported by a particular habitat.
- As the population approaches the carrying capacity of its habitat, the population growth curve takes the shape of an S (logistic).



Kurva pertumbuhan populasi (Spoolman and Miller, 2019)

Malthusian View – Population and Natural Resources

Natural resources have fixed limits (arithmetic/linear growth), while population growth tends to be exponential (geometric), which will ultimately lead to a crisis period that reduces the population back to levels that can be sustained by the environment.



- Thomas Robert Malthus

World Population Growth (1700 – 2022)

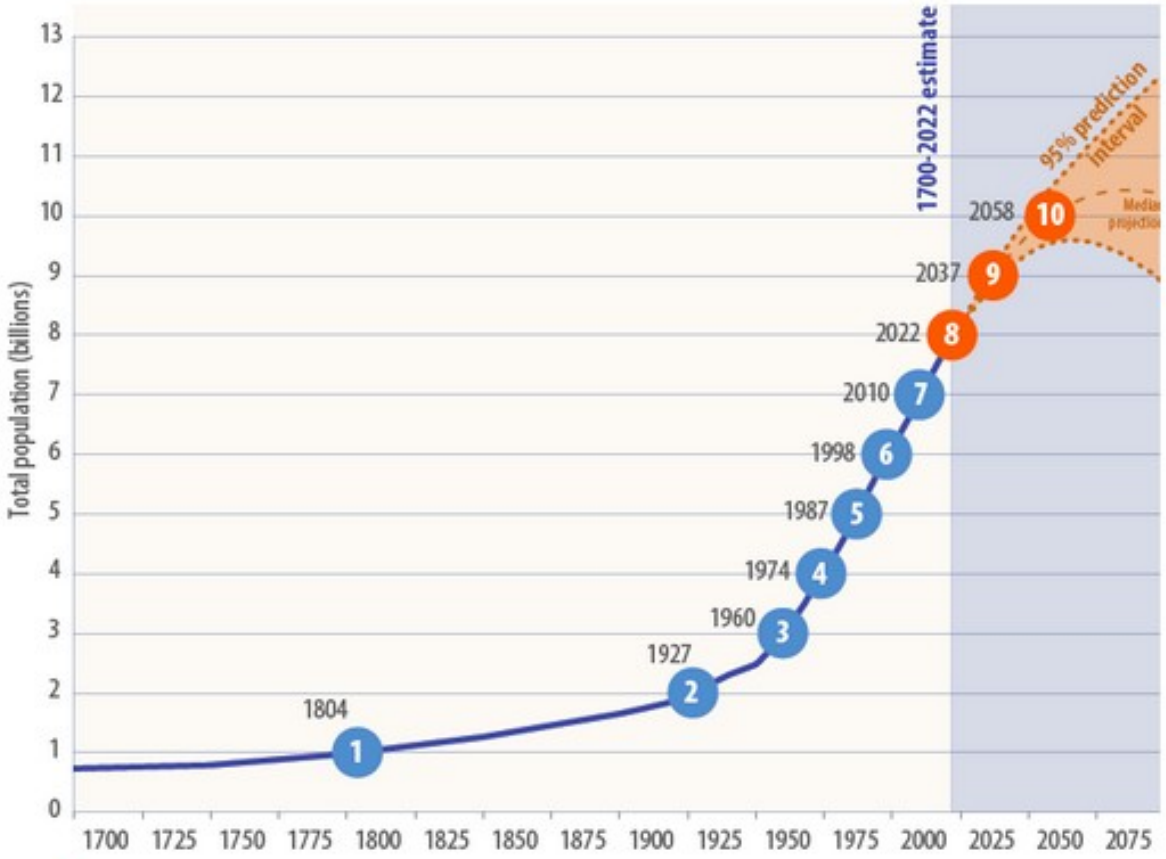
The human population has grown exponentially in recent centuries, Is this in line with Malthus's claims about the potential imbalance between population and resources?

What is the relationship between humans and the environment?

Can resource shortages be avoided? Do we have the ability to overcome it?

Figure 1

Global population size: estimates for 1700-2022 and projections for 2022-2100



Source: United Nations, DESA, Population Division (2022). World Population Prospects 2022.

Note: The solid blue line is the estimates from 1700 to today, the dotted red line the projection for the future up to 2100, and the dashed red line the upper and lower bounds of the 95% prediction interval for the projections.

Overpopulation Concept
When Polulation Growth become Overpopulation?

Population growth

The population increases over time.

Growth varies across regions.

Population pressure

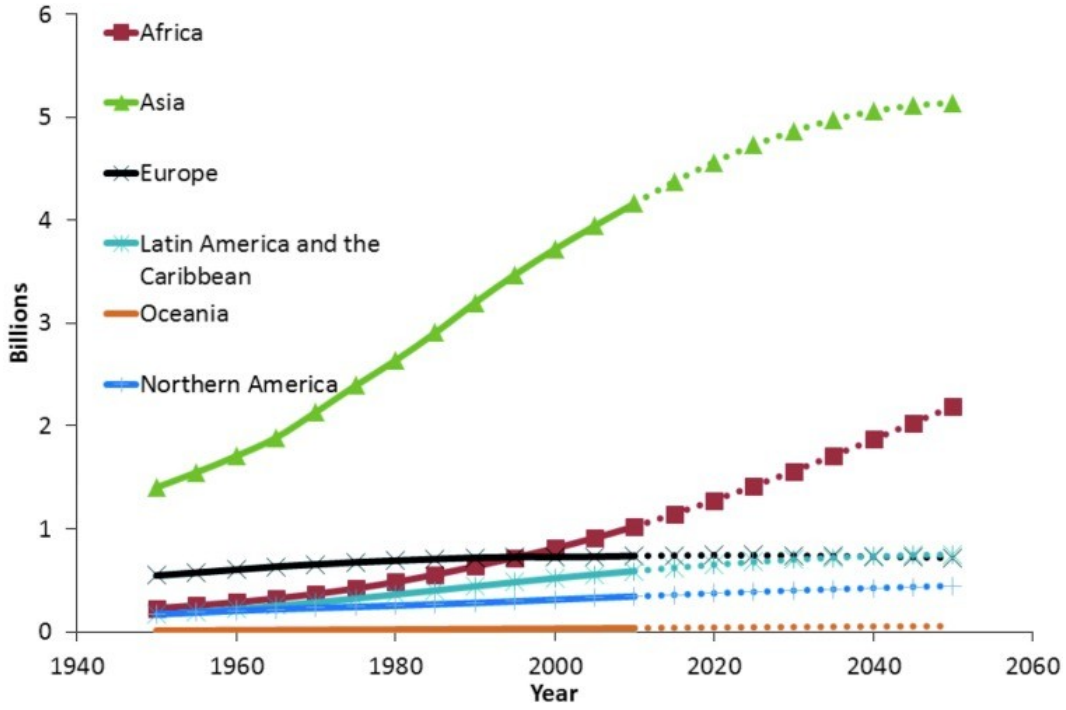
The demand for food, water, energy, land, housing, and infrastructure increases as the population grows.

Overpopulation

Occurs when the needs of the population sustainably exceed the carrying capacity of the system.

Not determined by population size alone.

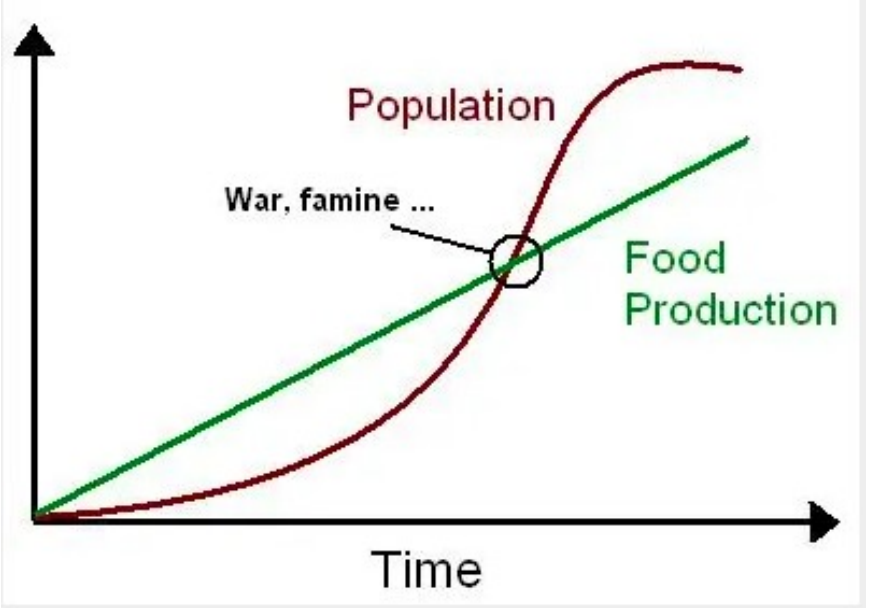
Also influenced by consumption levels, technology, resource distribution, and governance.



Population growth at different region

Neo-Malthusian Theorem

- Recognizing the interactions of social, economic, and environmental factors that lead to environmental damage and the scarcity of natural resources.
- However, this supports the premise: Natural resources are becoming scarcer, and environmental damage is occurring.



Neo-Malthusian Theory - The Role of Technology

Technology has a greater impact on the environment than population numbers.

Technologies such as petrochemical-based economies, the use of pesticides, fossil fuels, and other modern developments significantly increase the impact on the environment.

- Barry Commoner

Table 2.1 Who is overpopulated? Some comparisons of population, per capita gross domestic product, energy use, and other resource demands. Different places have widely divergent levels of population, affluence, and technology, with unclear implications for environmental impact.

Country	Total population (millions)[1]	GDP ($ per capita)[2]	Energy use (kg of oil equivalent per capita)[3]	Annual % total forest cover change (including plantations)[4]	Annual % forest cover change (natural forest only)[4]	Greenhouse gas emissions (tons of CO_2 equivalent per capita)[5]
China	1,294	3,936	861	+1.2	+0.6	3.91
USA	288	33,939	8,095	+0.2	+0.1	23.92
Bangladesh	143	1,527	133	+1.3	−0.8	0.38
Turkey	68	6,830	1,071	+0.2	. . .	4.07
UK	60	23,637	3,886	+0.6	+1.5	11.19
Kenya	32	1,003	489	−0.5	−0.5	0.81

1: 2002
2: 2000 (figures controlled for Purchasing Power Parity: equivalence of buying power in local markets)
3: 1999
4: 1990–2000
5: 2005
Source: Data from World Resources Institute Data (2005).

Country Figures (2021)						
Country	GDP (Million US$)	Rank	Energy Consumed (Exa Joules)	Rank	CO_2 Emission (Million Tonnes)	Rank
USA	23315.08	1	93.40	2	5007.34	2
China	17820.46	2	157.94	1	11472.37	1
Japan	5005.54	3	17.94	4	1067.40	4
Germany	4259.93	4	12.78	6	674.75	5
India	3150.31	5	34.51	3	2709.68	3
UK	3122.48	6	7.20	9	346.77	8
France	2957.88	7	9.40	8	305.96	10
Italy	2114.36	8	6.34	10	328.69	9
Canada	2001.49	9	13.84	5	545.63	7
South Korea	1810.96	10	12.56	7	616.08	6

Table-1: Comparison of GDP, Energy Consumption and CO_2 Emissions

Larger economies generally consume more energy and produce higher CO_2 emissions, but the relationship is not proportional.

Per Capita Figures (Year: 2021)			
Country	GDP (US$)	Energy Consumed (Joules)	CO_2 Emission (Tonnes)
USA	70219	277.20	14.86
China	12618	110.80	8.05
Japan	39827	144.00	8.57
Germany	51204	153.20	8.09
India	2238	24.50	1.93
UK	46586	107.00	5.15
France	43659	145.70	4.74
Italy	35770	107.00	5.55
Canada	52359	362.70	14.30
South Korea	34998	242.40	11.89

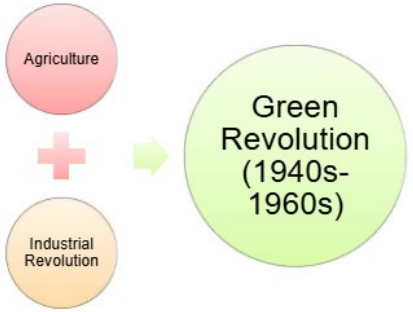
Higher GDP per capita tends to be associated with higher energy consumption, but CO_2 emissions also depend on efficiency and the energy sources used.

Table-2: Per Capita Figures of 2021 (GDP, Energy Consumption and CO_2 Emissions)

Agriculture Technology and Population Growth

- Humans do not rely solely on natural productivity →Agricultural innovation →Food fulfillment
- From agricultural extensification to intensification →Green revolution
- No longer just about Food **Sufficiency**, but **Food Security**
- *note: trade-offs* of intensification →pollution, GHG emissions, loss of biodiversity, etc







Agricultural technology helps increase the **effective carrying capacity** for human populations.

Limit to Population: An Effect rather than the Cause?

Economic Development ~poverty reduction ~ population decline

Or
Financial success as a 'prerequisite' for human development success

Graph →can occur

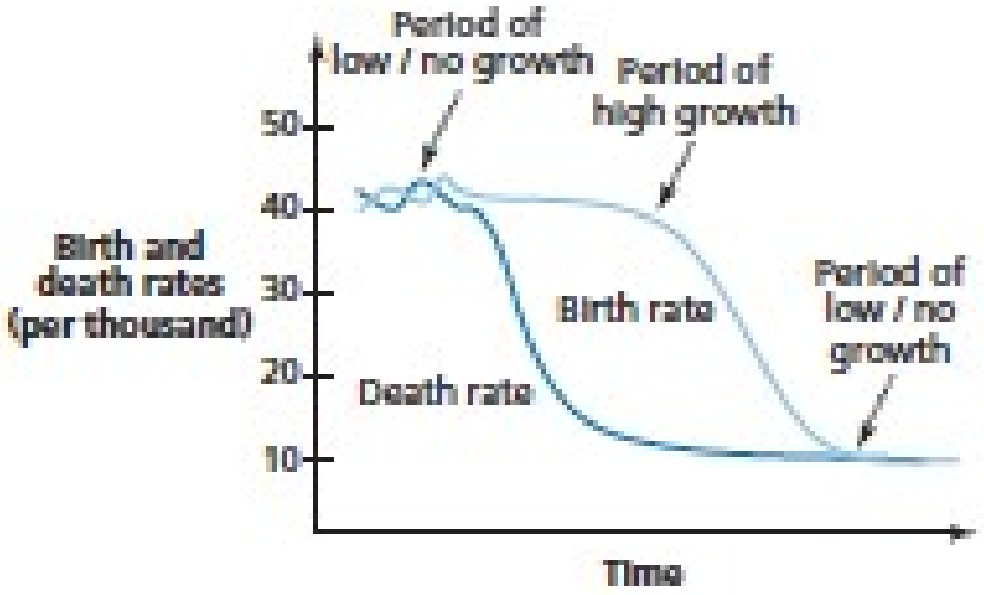


Figure 2.5 The demographic transition model. In theory, falling death rates lead to population growth, but as birth rates fall thereafter, the rate of growth slows, eventually ceasing when the two reach equilibrium.

Case Study : State of Kerala, India

Despite having a 'low' Per Capita Income Yet having human development (literacy rate, life expectancy, etc.) comparable to developed countries

2022

	Kerala	India	Australia	Jerman	AS	Indonesia
Population (milion)	36	1380	25,5	83,78	341	275
GDP (USD)	3100	2375	63945	48429	58487	3895
Population density (km2)	860	488	3,5	242	37,8	141
Literacy rate(%)	94	69,4	~100	~100	~100	94
Life expectancy (years)	75	67,3	83,1	80,5	77,5	68,3

Sumber World Population Data; Our World in Data;

Part 3.2 : Thinking About Environmental Impact

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From population to environment impact

- Previous discussions show that:
  - Population size affects total resource demand.

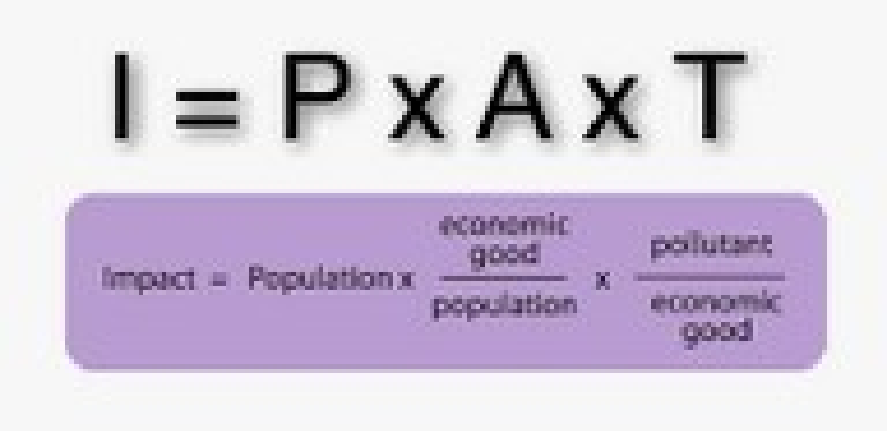
- However, countries with large populations do not always have the highest per capita energy consumption or emissions.

- The magnitude of impact is also influenced by:
  - consumption levels, and
  - how those needs are produced or met.
  - technology

Population, Development, and Environmental Impact

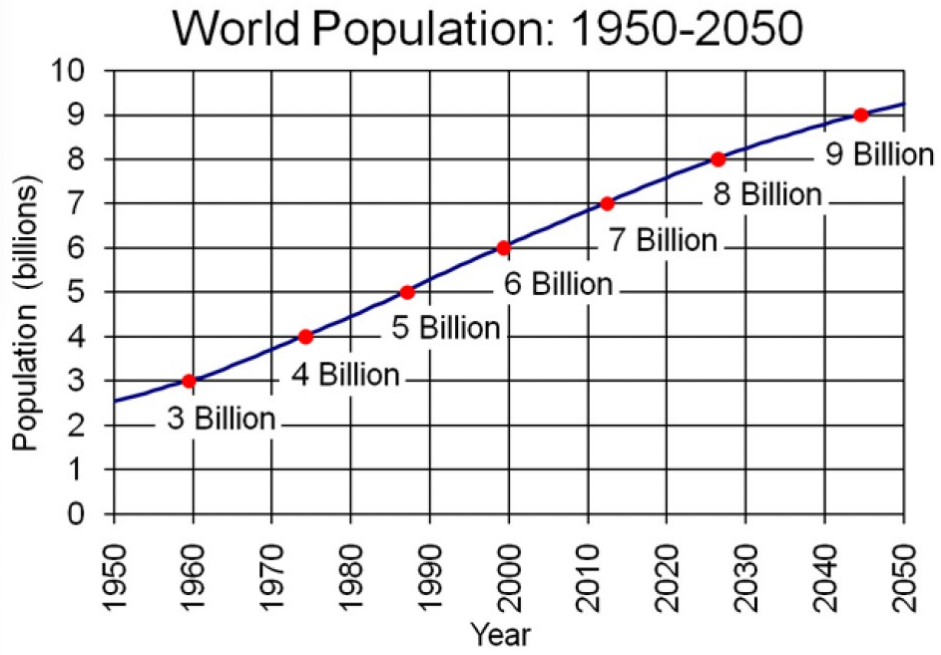
Erhlich & Holdren (1974) state that not only population but also the level of consumption impacts the environment.

The Level of Environmental Impact (I) is a result of Population (P), Affluence (A), and Technology (T).

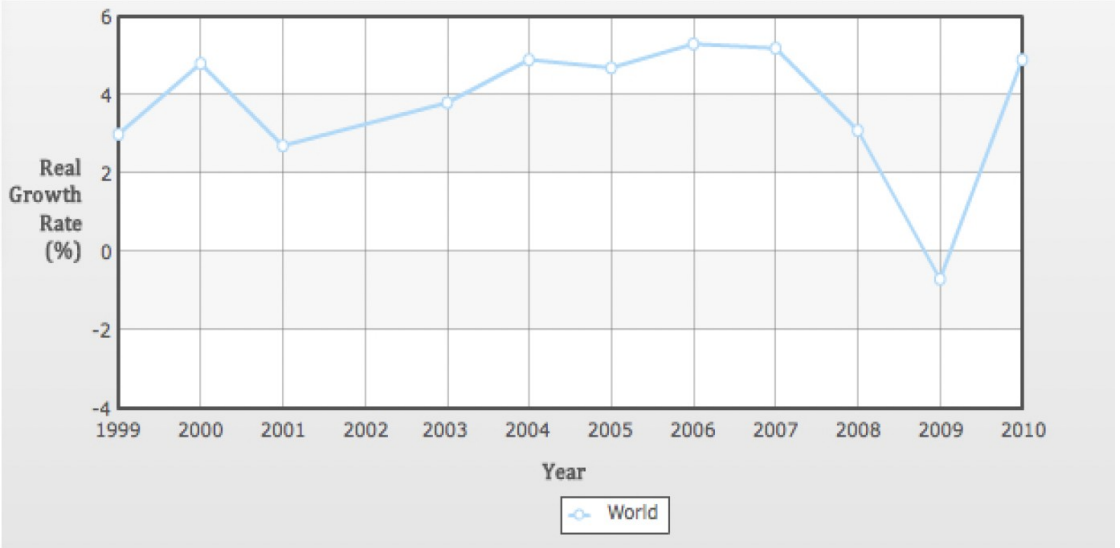


Affluence is measured by: The level of consumption of the population, or Per capita income.

P (opulation) and A(fluence) data



World Population Growth *Source: U.S. Census Bureau, International Data Base, December 2010 Update*



Worldwide Growth of Gross Domestic Product *Source: CIA World Factbook, Graph from IndexMundi*

How to maintain the Environmental Quality of 2050 like it is now?

Assumption: Technology Level Remains Constant (same as in 2010)

Meanwhile, the population in 2050 increases by 35%, and the Level of Affluence rises about 4 times,

The Technology Level must undergo 'decoupling' or reduce its impact by a factor of 'five'.

Equation:

$$P_{2010} \times A_{2010} \times T_{2010} = P_{2050} \times A_{2050} \times T_{2050}$$

or

Equation:

$$\frac{T_{2050}}{T_{2010}} = \frac{P_{2010}}{P_{2050}} \times \frac{A_{2010}}{A_{2050}} = \frac{1}{1.35} \times \frac{1}{4} = \frac{1}{5.4}$$

How to maintain environmental quality as technology advances?

Assumption: Technological Level Increases
Can solve environmental problems (Kuzweil, 1980) ~ Technofix



Coal-fired power plant technology generally increases GHG emissions, solar power reduces emissions.

Coal technology can reduce deforestation if it replaces the use of firewood for energy, cheap,

Dual Role of Technology

Improving Environmental Impact

- Efficiency in the Use and Management of Natural Resources
- Renewable Energy
- Pollution Monitoring
- Innovation

Worsening Environmental Impact

- Expansion of Natural Resource Exploitation
- Pollution and Waste
- Increasing Consumption

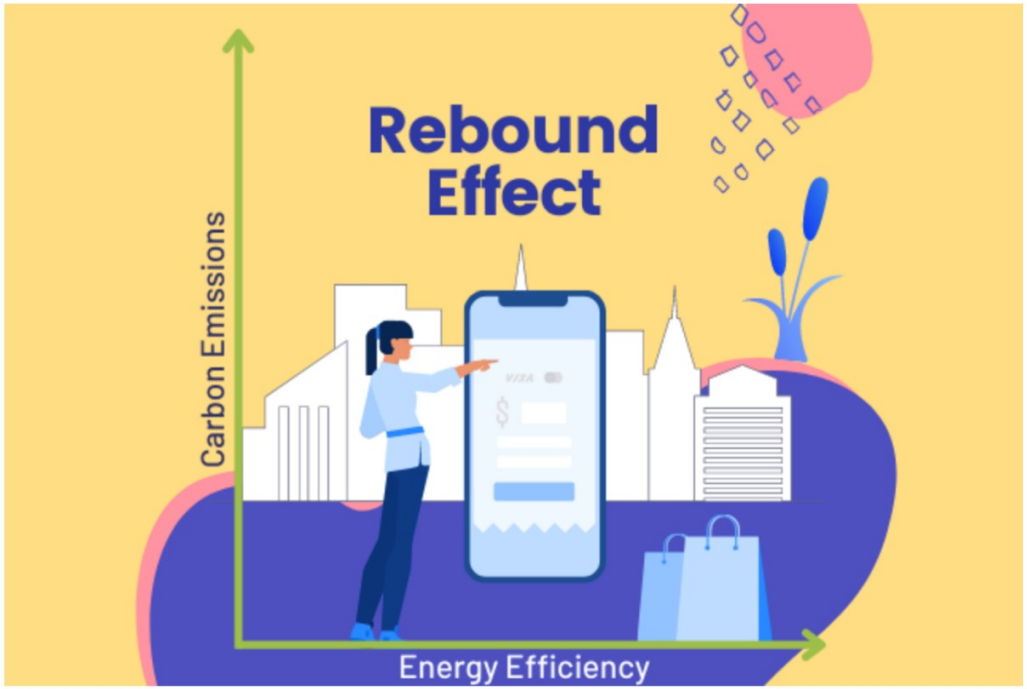
Technology, Consumption Patterns and Effects

Technological progress has made the use of natural resources more efficient, but it has also increased consumption of natural resources:

This is called the **'Jevons Paradox**'

Rebound: an increase in consumption impacts resource efficiency (RE = 1)

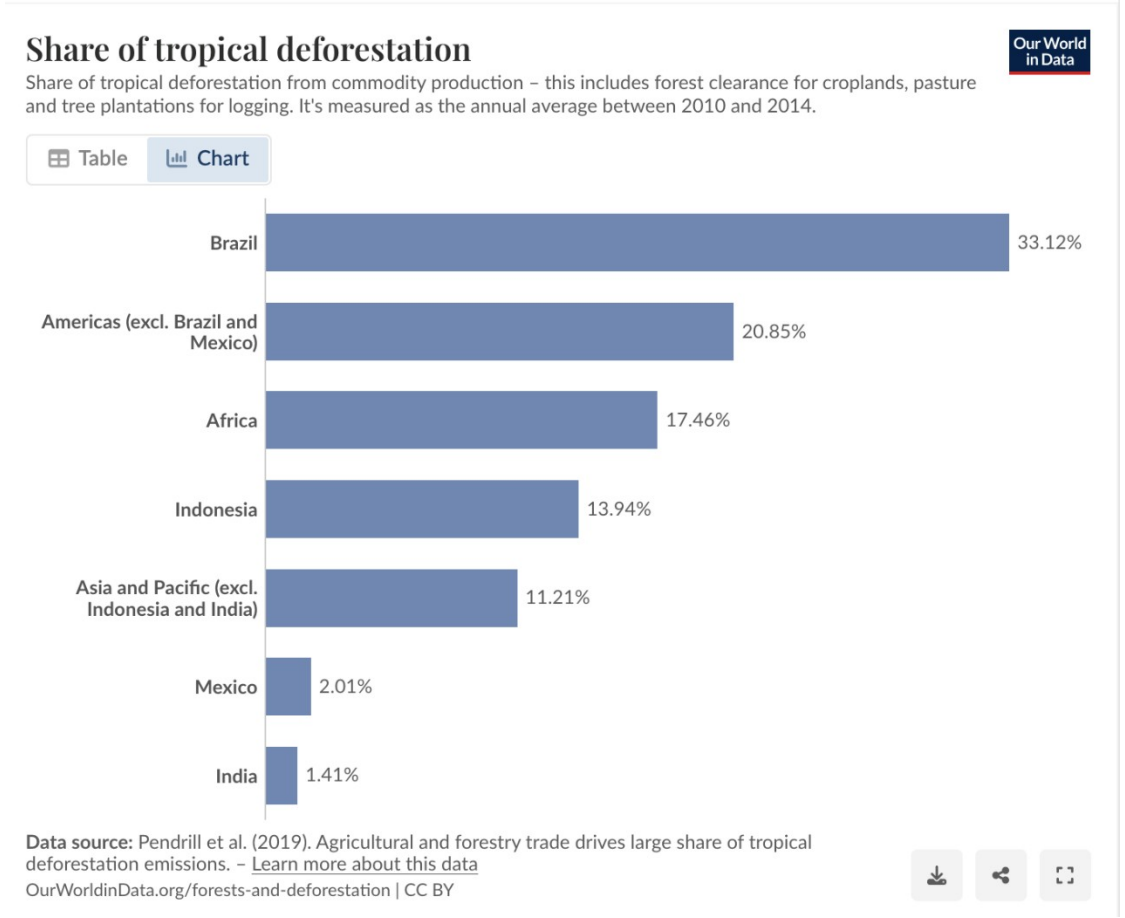
Backfire: Excessive consumption (Overconsumption) ~ RE < 1



Citation: Kaminski, J. (2023). Editorial. Reducing the Environmental Impacts of using Technologies. *Canadian Journal of Nursing Informatics, 18*(1). https://cjni.net/journal/?p=10851

What is the impact on the environment if 'Affluence' increases?

- According to Kuznets and Kurzweil, with increasing 'affluence' and more advanced technology, the impact on the environment declines.
- For example: reforestation has increased in Western European countries and the US, while it has decreased in Brazil and Indonesia.
- Is this argument by Kuznets and Kurzweil true?



Unequal Environmental Impact

Environmental impacts are not always borne by those who enjoy the benefits.

Goods consumed in one country can:

- use resources in another country,
- generate emissions and pollution in the producing country, while the benefits of consumption are enjoyed in the importing country.

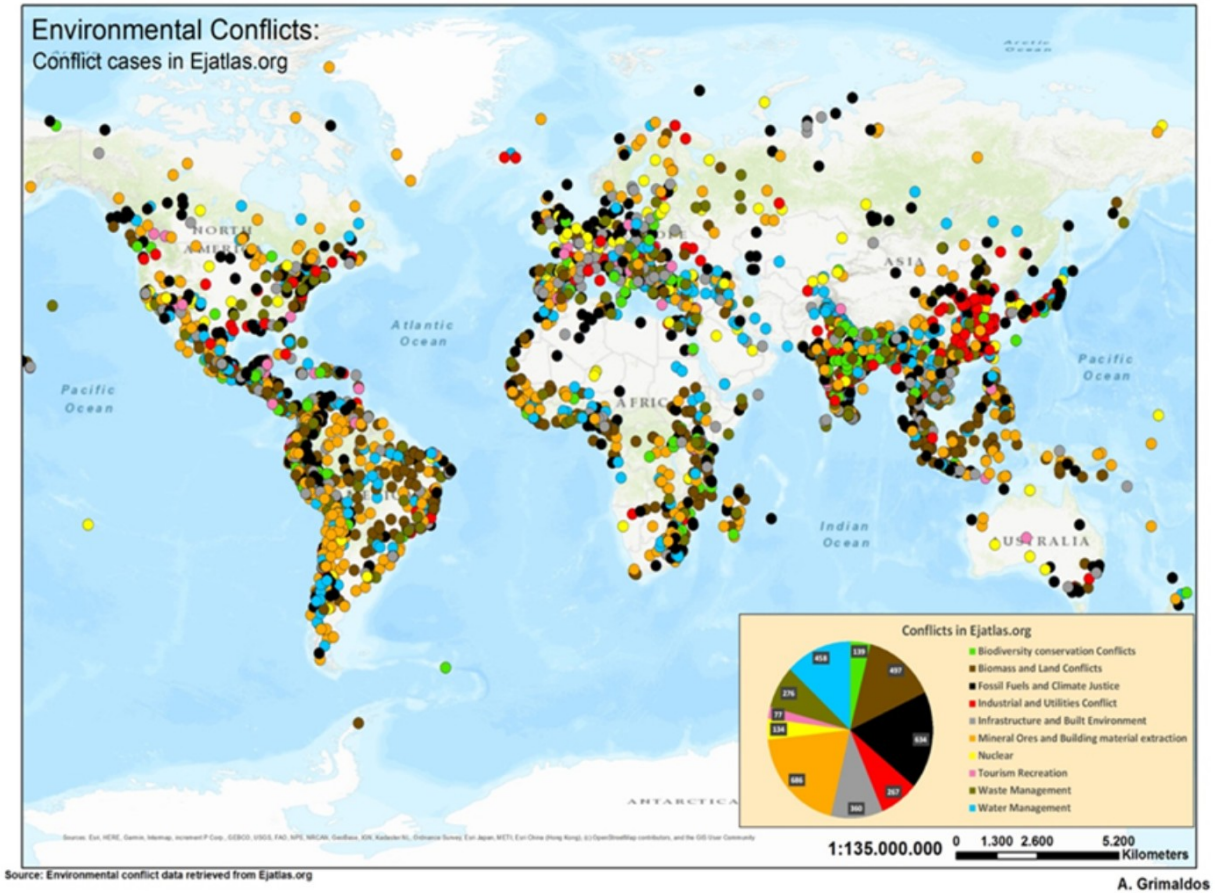
Examples:

- **Smartphones:** used in consumer countries, but their minerals may be mined in other countries.
- **Fast fashion:** clothing is consumed in one country, but water and energy usage as well as production waste occur in the producing region.
- **Agricultural commodities:** consumption in importing countries can be linked to land-use changes in producing countries.

Environment Justice

Offshoring: the practice of transnational corporations relocating production facilities;
Sourcing some raw materials from low- and middle-income countries (LMIC) (which often have looser environmental regulations)

Since the 1970s, many North American and European companies have relocated their production to Latin America and Asia, resulting in environmental externalities (environmental damage borne by other parties) (Robbins et al., 2014).



Map of environmental conflicts worldwide from https://ejatlas.org/ (Martinez-Alier, 2023)

In China, current electric vehicle (EV) policies can contribute significantly to climate change mitigation and air quality improvement, as well as provide public health benefits (Hsieh et al., 2022).

For electric vehicles (EVs), the demand for cobalt, lithium, manganese, and nickel by 2030 is projected to increase by 39.6, 19.6, 5.2, and 4.7 times, respectively (compared to 2015) (Jones et al., 2020)—much of which originates from developing regions.

Severe environmental impacts occur at the mineral extraction stage, such as local groundwater depletion during lithium mining in Chile, as well as social issues like unsafe working conditions and child labor in the mines of the Democratic Republic of the Congo (Prates et al., 2023).

Environment Justice–*EV Battery (LFP* and *NMC)*

There are two types of batteries for EVs:

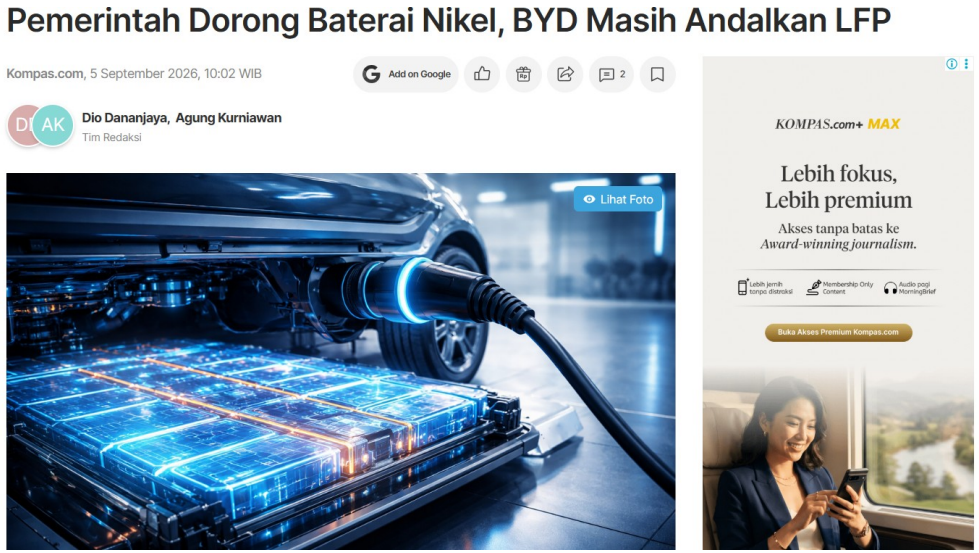
LFP – Lithium-Iron-Phosphate

NMC – Nickel-Manganese-Cobalt

The EV industry's response to reducing environmental impact is by using LFP batteries due to their lower environmental footprint.

In China, 70% of EVs use LFP.

How abour Indonesia?



Pemerintah Dorong Baterai Nikel, BYD Masih Andalkan LFP

Kompas.com, 5 September 2026, 10:02 WIB

Add on Google

Dio Dananjaya, Agung Kurniawan

Tim Redaksi

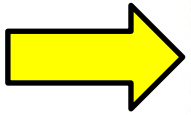
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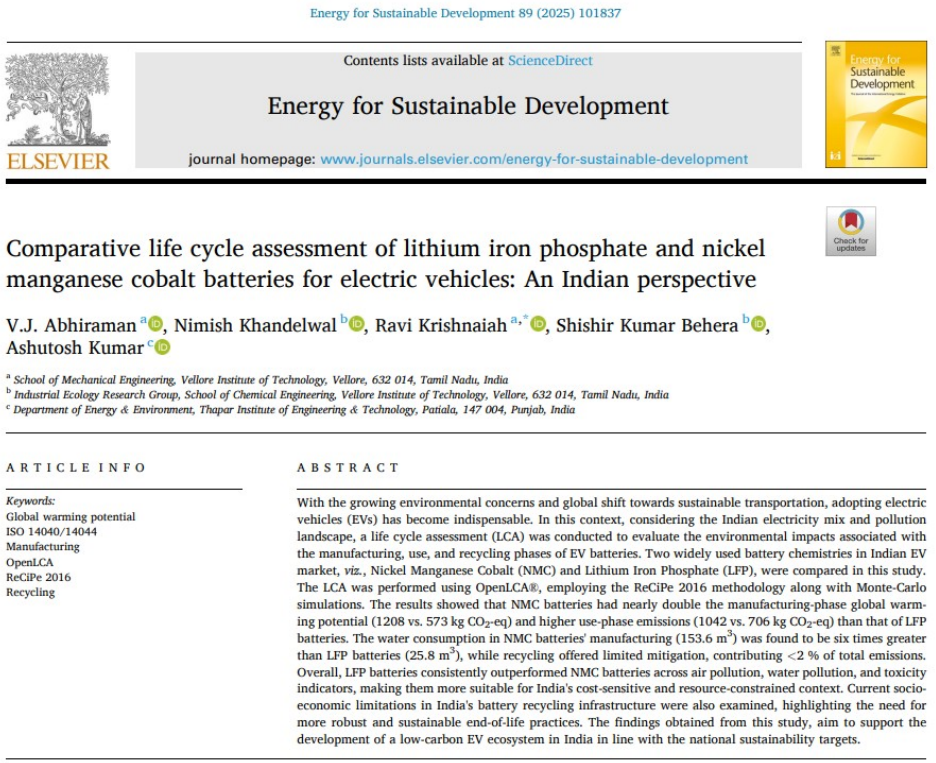
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Comparative life cycle assessment of lithium iron phosphate and nickel manganese cobalt batteries for electric vehicles: An Indian perspective

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ABSTRACT

With the growing environmental concerns and global shift towards sustainable transportation, adopting electric vehicles (EVs) has become indispensable. In this context, considering the Indian electricity mix and pollution landscape, a life cycle assessment (LCA) was conducted to evaluate the environmental impacts associated with the manufacturing, use, and recycling phases of EV batteries. Two widely used battery chemistries in Indian EV market, *viz.*, Nickel Manganese Cobalt (NMC) and Lithium Iron Phosphate (LFP), were compared in this study. The LCA was performed using OpenLCA®, employing the ReCiPe 2016 methodology along with Monte-Carlo simulations. The results showed that NMC batteries had nearly double the manufacturing-phase global warming potential (1208 vs. 573 kg CO_2-eq) and higher use-phase emissions (1042 vs. 706 kg CO_2-eq) than that of LFP batteries. The water consumption in NMC batteries' manufacturing (153.6 m^3) was found to be six times greater than LFP batteries (25.8 m^3), while recycling offered limited mitigation, contributing <2 % of total emissions. Overall, LFP batteries consistently outperformed NMC batteries across air pollution, water pollution, and toxicity indicators, making them more suitable for India's cost-sensitive and resource-constrained context. Current socio-economic limitations in India's battery recycling infrastructure were also examined, highlighting the need for more robust and sustainable end-of-life practices. The findings obtained from this study, aim to support the development of a low-carbon EV ecosystem in India in line with the national sustainability targets.

GWP of NMC production is around **1,208 kg** CO_2**-eq**, while LFP is **573 kg** CO_2**-eq**.

Water consumption in NMC production is approximately **153.6** m^3, compared to LFP at **25.8** m^3.

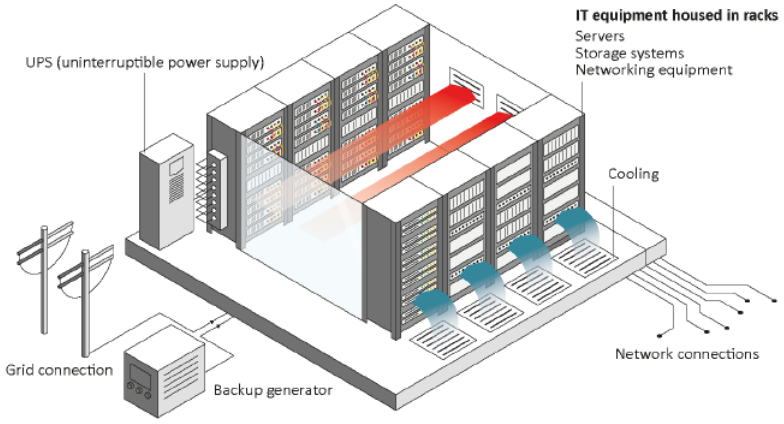
LFP also performs better across several air pollution, water pollution, and toxicity indicators.

Data Center for AI

AI requires **data centers that demand a significant amount of electrical energy**. Data centers consumed around **415 TWh of electricity in 2024**, or **1.5% of global electricity consumption**.
Demand is projected to rise to approximately **945 TWh by 2030**, with AI as the primary driver.

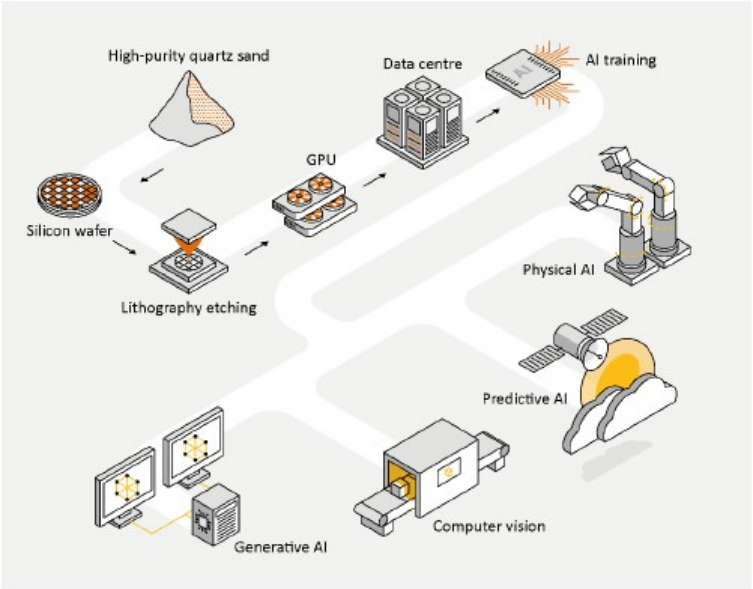
Note: While the benefits of AI are enjoyed by many, its energy and water demands are concentrated in the regions where data centers are built—are these impacts distributed equitably?

Figure 2.1 ▷ Data centre components



IEA. CC BY 4.0.

Effective design and integration of key data centre components ensure continuous operation and optimal performance



IEA. CC BY 4.0.

AI is supported by a highly complex global supply chain





IEA. CC BY 4.0.

Data centres are often located in large clusters, potentially creating challenges for local electricity systems

Notes: GW = gigawatt. We define a data centre cluster as a group of data centres located within 100 km of each other. The ten largest clusters have been named. The Pearl River Delta encompasses the combined capacity of Guangzhou, Shenzhen and Hong Kong (China).

Source: IEA analysis based on data from OMDIA (2025).

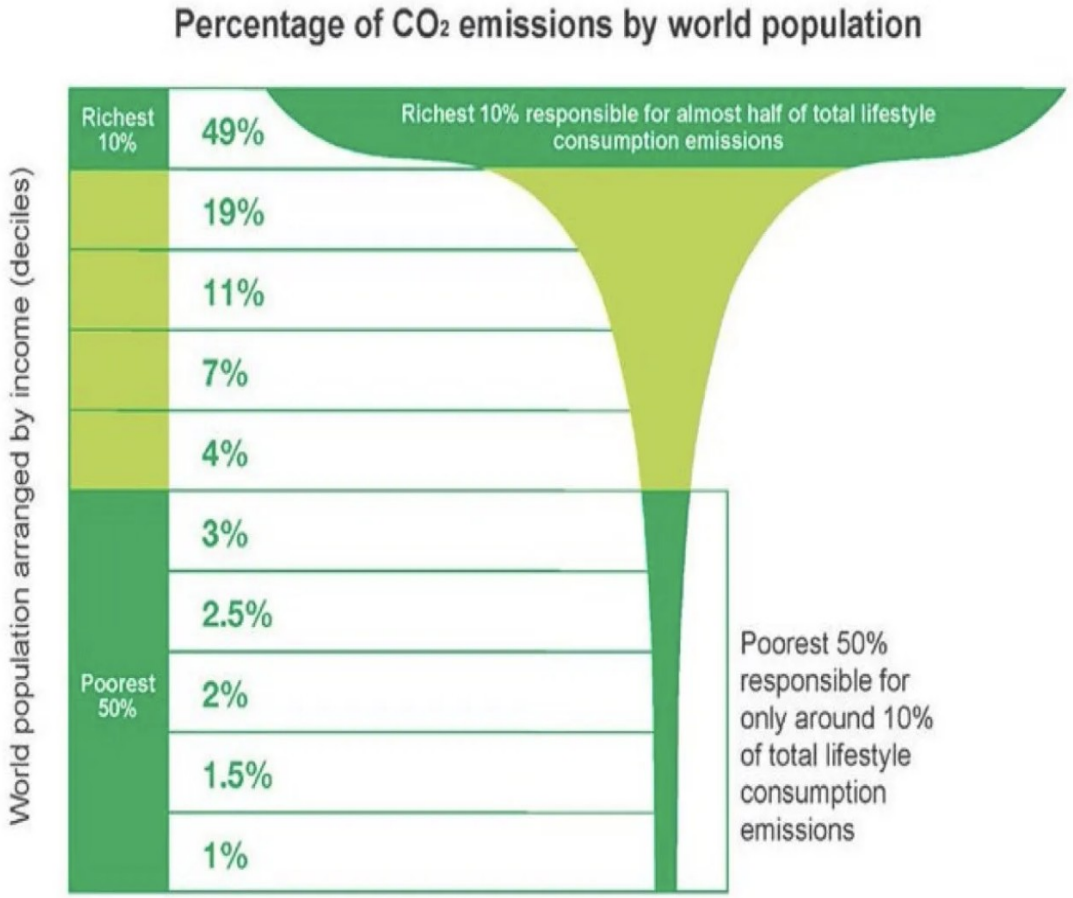
Part 3.3 :
Over-Consumption

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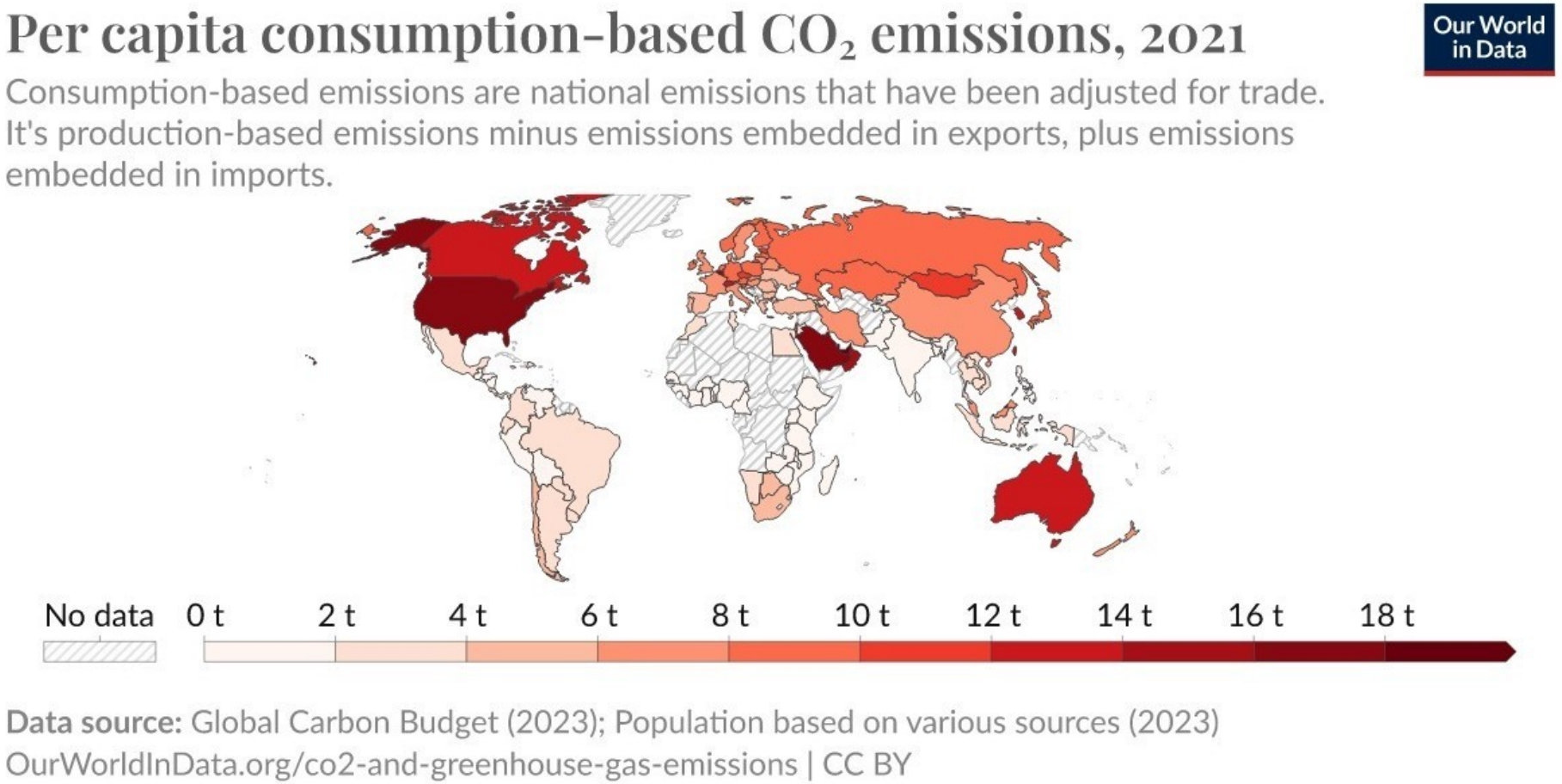


Over-Consumption Concept

- The level of excessive consumption that leads to (drivers of) environmental damage
- The perspective on consumption behavior: rational, cultural, pleasurable, identity, and communication



CO_2 emissions per capita based on consumption (2021) by country



Overconsumption – Addiction

Becoming addictive is not related to actual needs:

- Addictive Consumption: results in low levels of individual health and participation in society.
- Example: obesity.

- (John Ehrenfeld)



Overconsumption

Overconsumption

- *Overconsumption is not always related to economic capability.*
- *The comparison between the Global South (export of natural resources, rapid urbanization) and the North (consumptive lifestyle, high industrialization)*

Overconsumption example:

- The use of air conditioning at very low temperatures throughout the day,
- turning on lights when not in use.
- Eating portions larger than necessary and leaving leftovers.
- Single-use packaging (use of plastic and disposable utensils)
- Overuse of fertilizers beyond the plants' needs..







Overconsumption – Hyperconsumption

- Currently, geographical and time boundaries are increasingly negligible; the use of technology is becoming more entrenched in daily life.
- Using the latest / forefront / most up-to-date ~ consumption.

(Giles Lipovetsky, 2011)

- **Overconsumption**: using resources more than needed or exceeding the ecosystem's ability to recover → degrading environmental quality.

- **Hyperconsumption**: purchasing/using driven by social/psychological factors far beyond functional needs (speed, novelty, status), with rapid turnover and many duplications.

Hyperconsumption

- Hyperconsumption is driven by social and psychological factors.
- Trends and technological advancements influence the occurrence of hyperconsumption.
- For example: Loan modes (SpayLater, credit, installment schemes) make it easier for consumers to purchase goods.



Example:

- *Fast fashion: buying the latest clothing styles every month.*
- *Fast furniture: the production and consumption of furniture (usually low priced with low durability).*
- *Replacing gadgets every year for the latest models..*



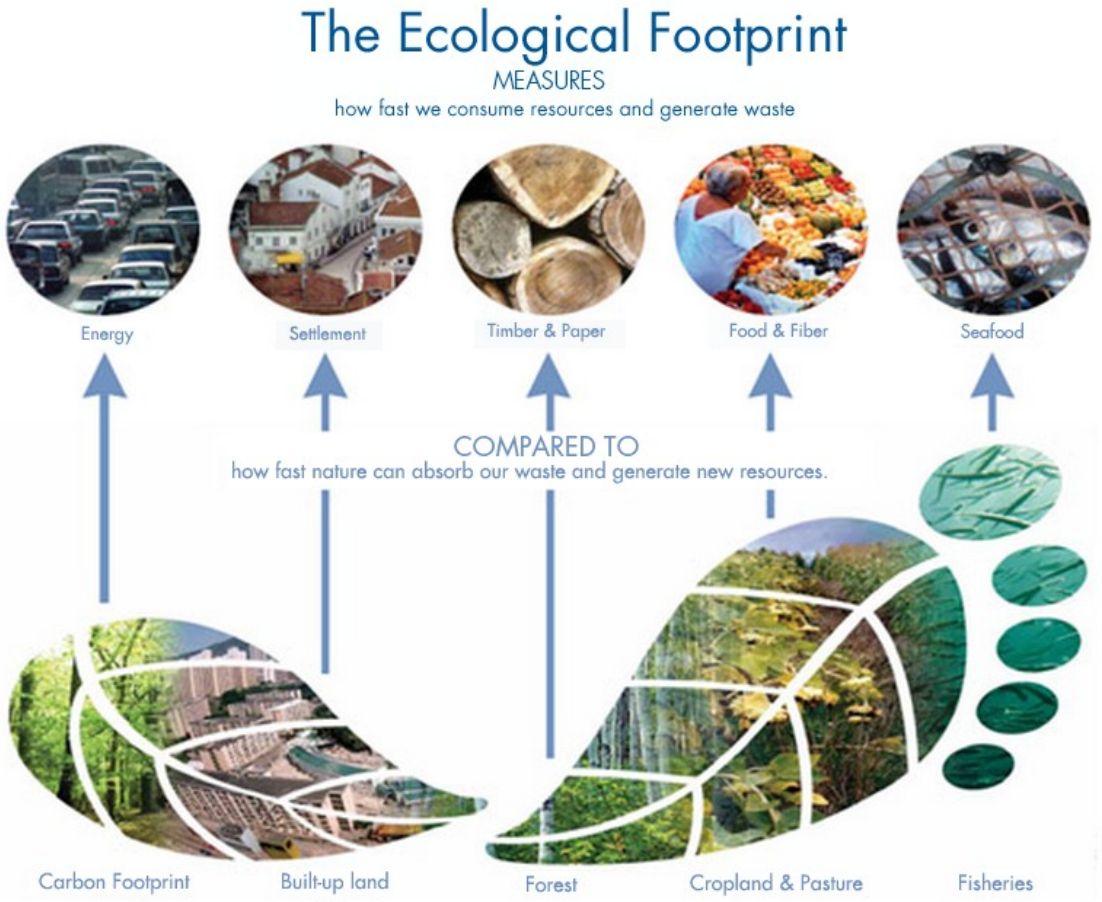
Part 3.4 :
Ecological Footprint



WI 1103 Introduction to Principle of Sustainability

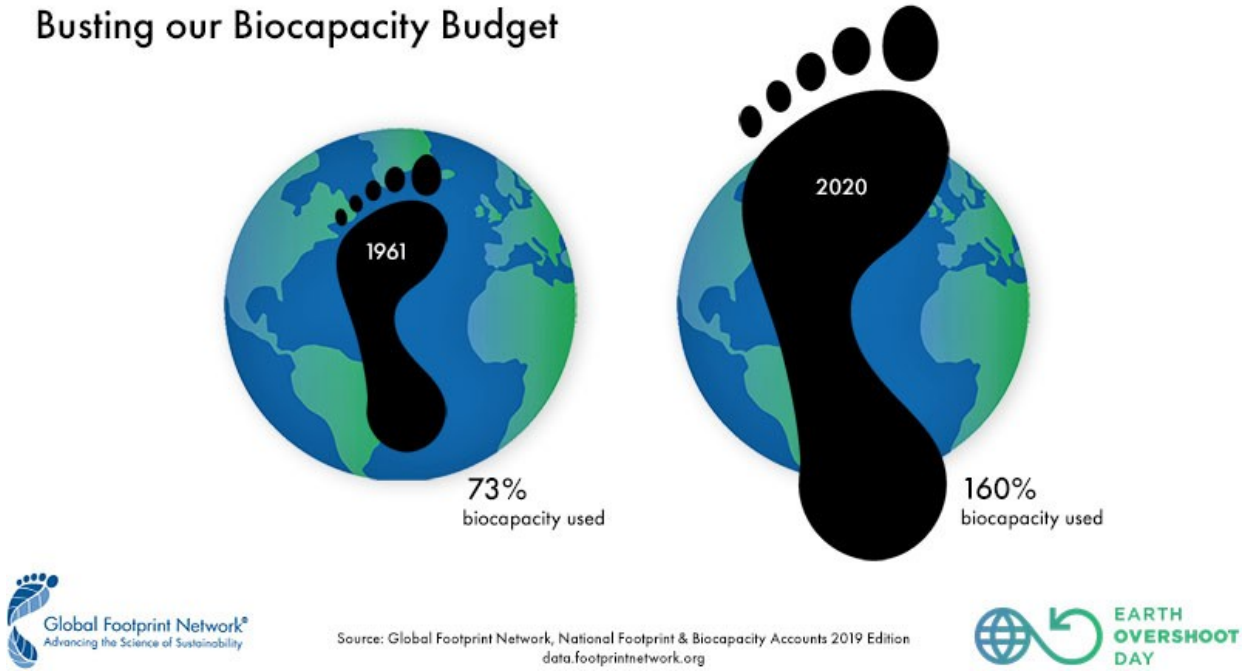
Ecological footprint: Concept

- is a measure of human demand for natural capital (= the amount of nature needed to support humans or an economy) → in other words, a measure of human impact on the environment.
- This demand is tracked through an ecological accounting system that measures the land area required to provide resources for human activities and to absorb its waste.
- The more resources consumed by an individual or a society, the larger its ecological footprint.



https://www.footprintnetwork.org/our-work/ecological-footprint/

- The calculation compares the ECOLOGICAL FOOTPRINT (the area of productive biological land needed to supply everything consumed by humans) with BIOCAPACITY (biocapacity = the area of productive biological land available to produce what humans demand from nature).

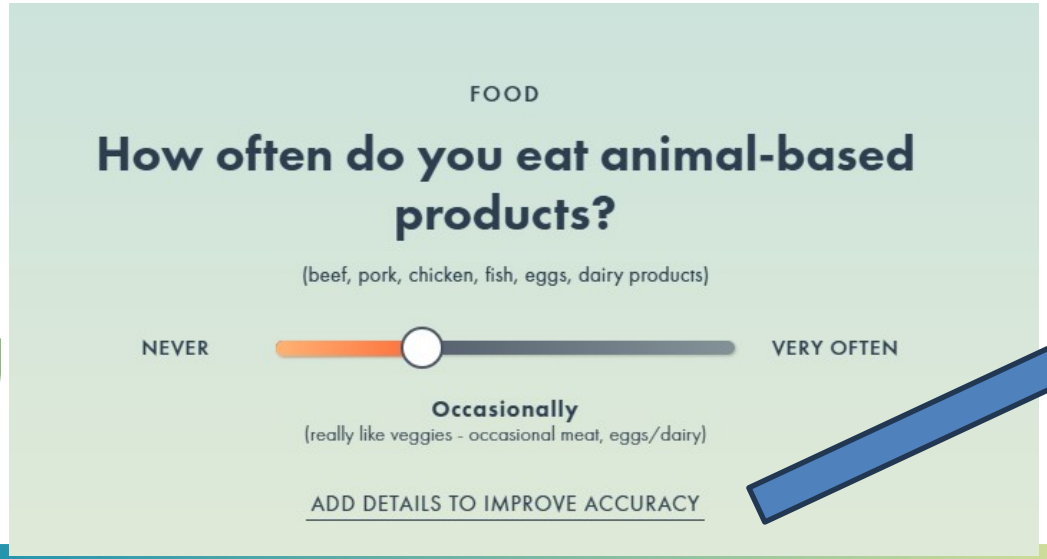


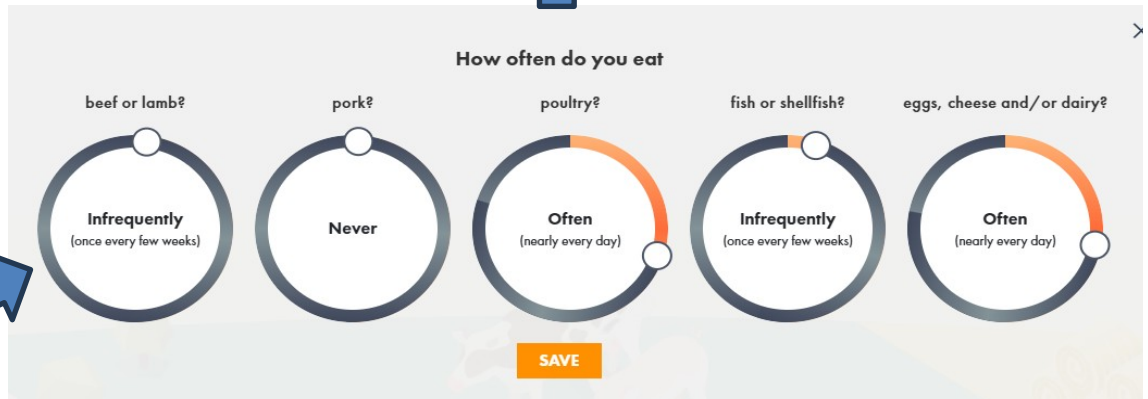
https://overshoot.footprintnetwork.org/content/uploads/2020/08/footprint_Earth_bc_1961_2020_small.jpg

Ecological footprint

- Measurement at the individual level
- Main unit: global hectare (gHA) per person per year—the average productive land/sea area 'required' by our lifestyle.
  - Food consumption
  - Energy and material use for homes
  - Mobility (use of transportation and traveling)
  - Waste produced







https://www.footprintnetwork.org

X INDONESIA (2022)

GDP PER PERSON
$4,788

POPULATION
275,500,992

Biocapacity per person
1.2
gha

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Ecological Footprint per person
1.8
gha

=

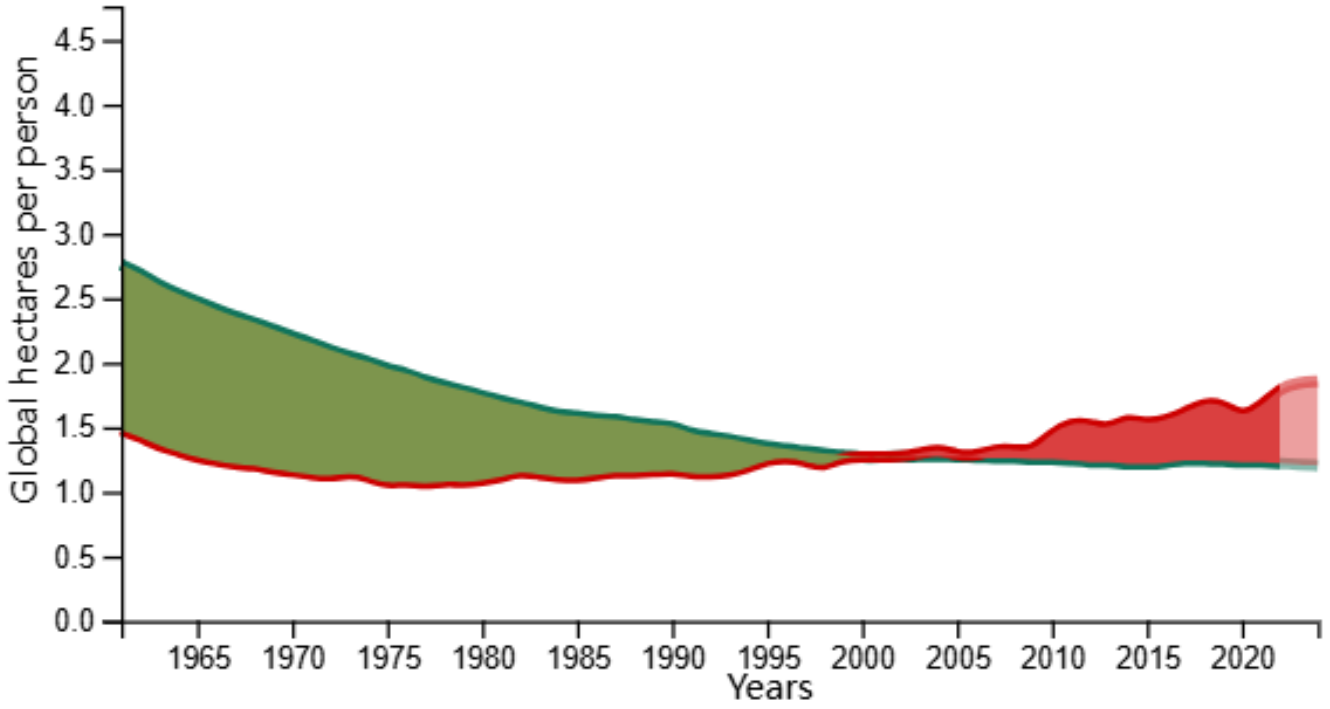
BIOCAPACITY RESERVE(+)/DEFICIT(-)
-0.6
gha

Ecological Footprint and Biocapacity
From 1961 to 2024
(last 2 years are estimates)

Ecological Footprint per person

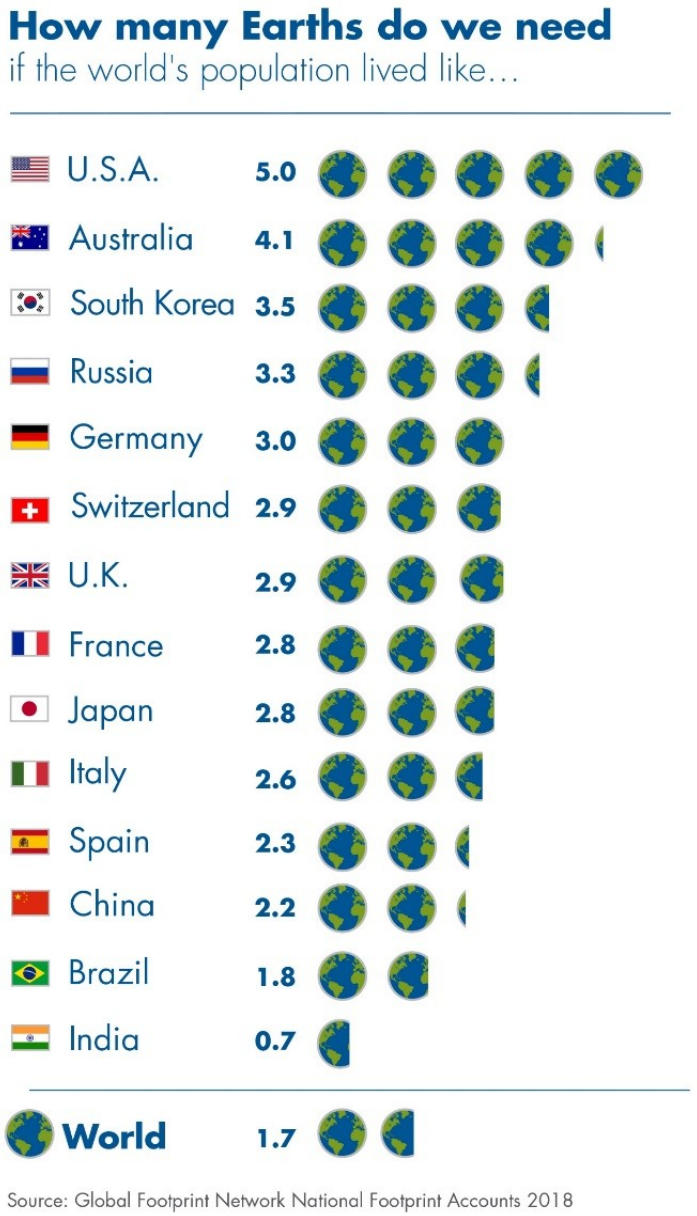
Biocapacity per person

Learn More

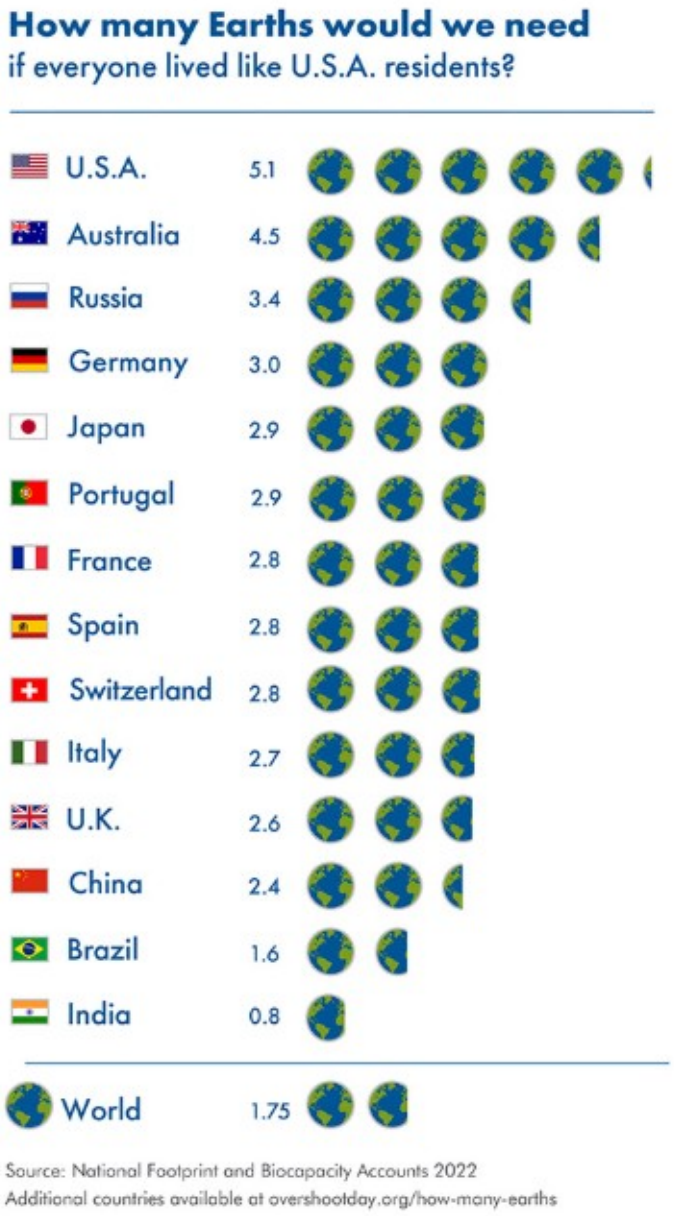


How many Earths would be needed if the population lived like …

2018



2022



Discusion

- Use each individual's Ecological footprint calculator. What results did you get?
  - Ecological footprint calculator: https://www.footprintcalculator.org/home/en

- What can you do to change each individual's Ecological footprint? Link the changes made to reduce excessive consumption, whether from changes in consumption patterns or can be aided by technology.

- Can all changes be made individually? Determine whether the changes can be made at the individual level, or must they be collective or through government or global policies?

Thank you